

SHUTIAN LUO

CONTACT

Assistant Professor
Department of Computer Science, University of Macau
Location: Macau SAR, China

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[Homepage](#) and [Google Scholar](#)

EDUCATION

University of Chinese Academy of Sciences Sept. 2016 - Jan. 2023
Doctor of Philosophy, Computer Applied Technology
Dissertation Advisor: Prof. [Chengzhong Xu](#). Thesis: Research on Key Technologies of Intelligent Resource Management for Cloud Native Microservices Workload.

Shenzhen University Sept. 2012 - Jul. 2016
Bachelor of Engineering, Electronic Science and Technology
Postgraduate Recommendation (**3 out of 283**)

RESEARCH INTERESTS

- AI systems and infrastructure for scalable foundation models and generative AI
- GPU- and CPU-GPU-heterogeneous systems for large-scale LLM and MoE serving

WORK EXPERIENCE

Assistant Professor, University of Macau Aug. 2026 - Present
Department of Computer Science

- Lead the AI Infra Group, building efficient AI infrastructure over emerging heterogeneous computing architectures, with a current focus on efficient LLM systems, CPU-GPU superchips, and physical AI.

Senior Research Scientist, ByteDance Seattle Dec. 2025 - July. 2026

- Designed and implemented asymmetric GEMM kernels for next-generation NVIDIA Blackwell superchips, optimized for heterogeneous memory and high-bandwidth CPU-GPU interconnects.

Postdoctoral Researcher, University of Virginia Feb. 2025 - Dec. 2025
Advisor: Prof. [Haiying Shen](#)

- Built NVIDIA Superchip inference systems: zero-copy KV-cache offloading and on-demand MoE expert streaming, improving long-context latency, memory efficiency, and throughput.

Postdoctoral Researcher, Yale University Jun. 2023 - Jan. 2025
Advisor: Prof. [Lin Zhong](#) and Co-advisor: Prof. [Anurag Khandelwal](#)

- Explored the trade-off between performance and programming transparency for data-parallel workloads in RDMA-based distributed shared memory (DSM) systems.

Research Assistant, University of Macau Apr. 2021 - May 2023
[Cloud and Distributed Systems Lab](#)

- Investigated workload variability in microservices using deep learning for predictive autoscaling, and proposed a resource management framework to ensure SLA requirement in shared environments.

Research Intern, Alibaba Cloud Nov. 2018 - Jan. 2023

- Designed resource management system for online shopping application in Alibaba to handle extremely high load during Double 11 Festival for high performance.

SELECTED PUBLICATIONS

1. [OSDI'26] Shutian Luo and Haiying Shen. No Buffer, No Bottleneck: Efficient Zero-Copy KV Cache Offloading for Long-Context LLMs. In Proceedings of the 20th USENIX Symposium on Operating Systems Design and Implementation. 2026.
2. [HPDC'26] Liao Chen, Chenyu Lin, Junlin Chen, Shutian Luo, Huanle Xu and Chengzhong Xu. Cremes: Cost-Efficient and Reliable Microservice Execution on Spot Instances. In Proceedings of the 35th ACM International Symposium on High-Performance Parallel and Distributed Computing. 2026.
3. [ASPLOS'25] Shutian Luo, Jianxiong Liao, Huanle Xu, Zhi Zhou and Chengzhong Xu. Embracing Imbalance: Dynamic Load Shifting among Microservice Containers in Shared Clusters. In Proceedings of the 30th ACM International Conference on Architectural Support for Programming Languages and Operating Systems. 2025.
4. [SoCC'25] Yanying Lin, Shuaipeng Wu, Shutian Luo, and Hong Xu, Haiying Shen, Chong Ma, Min Shen, Le Chen, Chengzhong Xu, Lin Qu, Kejiang Ye. Understanding Diffusion Model Serving in Production: A Top-Down Analysis of Workload, Scheduling, and Resource Efficiency. In Proceedings of the 16th ACM Symposium on Cloud Computing. 2025.
5. [ISCA'24] Liao Chen*, Shutian Luo*, Chenyu Lin, Zizhao Mo, Huanle Xu, Kejiang Ye and Chengzhong Xu. Derm: SLA-aware Resource Management for Highly Dynamic Microservices. In Proceedings of the 51th International Symposium on Computer Architecture. 2024.
6. [ASPLOS'23] Shutian Luo, Huanle Xu, Kejiang Ye, Guoyao Xu, Liping Zhang, Jian He, Guodong Yang, and Chengzhong Xu. Erms: Efficient Resource Management for Shared Microservices with SLA Guarantees. In Proceedings of the 28th ACM International Conference on Architectural Support for Programming Languages and Operating Systems. 2023.
7. [ACM SoCC'21] Shutian Luo, Huanle Xu, Chengzhi Lu, Kejiang Ye, Guoyao Xu, Liping Zhang, Yu Ding, Jian He, and Chengzhong Xu. Characterizing Microservice Dependency and Performance: Alibaba Trace Analysis. In Proceedings of the 12th ACM Symposium on Cloud Computing. 2021. (**Best Paper Award. The first paper in Asia.**)
8. [ACM TOCS] Shutian Luo, Chenyu Lin, Kejiang Ye, Guoyao Xu, Liping Zhang, Guodong Yang, Huanle Xu and Chengzhong Xu. Optimizing resource management for shared microservices: a scalable system design. In Proceedings of ACM Transactions on Computer Systems. 2024. (**The 10th paper from China in the past 40 years.**)

HONORS/ AWARDS

- ACM SoCC'2021 **Best Paper** Award.
- 2024 CCF Computer Architecture Outstanding Ph.D (**5 awarded annually**).
- 2024 IEEE Technical Community on Scalable Computing (TCSC) Outstanding Ph.D.
- 2024 Chinese Academy of Sciences Outstanding Ph.D. (**the first Ph.D.** to receive this award in the past 16 years at Shenzhen Institute of Advanced Technology, Chinese Academy of Sciences).
- **Alibaba Star offer** (P7 level, highly professional expert). Alibaba Star is the highest offer for Ph.D graduates, with no more than 10 positions offered annually at Alibaba.
- Outstanding Science Research Intern at Alibaba Cloud in 2021 (**awarded to only 12 interns**).

OPEN-SOURCED PROJECTS

- Released two versions of Alibaba microservices traces: 100GB v2021[[Github](#)] and 2TB v2022[[Github](#)]
- Designed an efficient resource management System for microservices [[Github](#)], earning three ACM badges for availability, functionality, and reproducibility from the conference committee.

TEACHING

- Instructor, Distributed Computer Systems (CISC3010), University of Macau.

- Guest Lecturer, Cloud-Native Computing and Microservices, University of Virginia, 2025.

COMMUNITY SERVICES

- PC member for EuroSys 2027.
- TPC member for TPDS, COINS 2025, SmartData-2024, HDIS 2023, IEEE ScalCom 2022.